

PRIORITIZING OPERATIONAL READINESS FOR AIRPORTS

Examining the benefits of implementing an ORAT framework early for new development and terminal expansion

This analysis builds upon WSP's ORAT article (published in 2024), which raised the importance of an early start to ORAT. Here, Tara Durnin, Aviation Director, Middle East, looks closely at the value of introducing the ORAT framework at the planning stage, exploring ORAT as a strategic investment that directly impacts an airport's financial stability, operational efficiency and long-term success.

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Few industries face as much complexity and pressure in the delivery of infrastructure as airports. Every decision made during the planning and design phases of an airport project carries long-term implications for operations, passenger experience and financial performance. Yet, many projects remain overly focused on the construction phase, treating operational readiness as an afterthought. The result has been costly delays, chaotic openings and long-term reputational damage. As airports continue to grow, evolve and modernize, ORAT[1] is becoming critical in advancing trouble-free transitions from construction to full operation. This article examines why airports should establish ORAT upfront at the planning stage and how early implementation benefits both airports and the aviation sector.

Begin with the End in Mind

In airport projects, the principle *begin with the end in mind* is critical to understanding all aspects that affect the achievement of an outstanding airport opening.

A poorly executed airport opening can result in baggage mishandling, flight delays, security lapses and passenger dissatisfaction. ORAT mitigates these risks by proactively identifying and resolving issues before day one of operations. By envisioning the airport opening-day strategy from the very start, every decision, from design to construction, can be aligned with these end goals.

A well-executed ORAT program means a fluid and hassle-free process for passengers, contributing to a strong first impression and positive word-of-mouth that can boost the airport's reputation. Retail, concessions and airline partnerships play a significant role in revenue generation for airports. ORAT supports a high-performance start, reducing revenue losses associated with operational inefficiencies.

Key Aspects of an ORAT Program:

- **Stakeholder Coordination & Governance**: Engaging airlines, operators, regulators, and service providers early ensures seamless collaboration and alignment with operational needs.
- **Operational Concept & Design Alignment**: Operational requirements should guide facility design, optimizing passenger flow, baggage handling, and security to prevent costly modifications later.
- **Standard Operating Procedures (SOPs) & Staff Training**: Clearly defined procedures and comprehensive training programs equip personnel to handle new systems and workflows efficiently from day one.
- **Systems Integration & Testing**: All technological and operational systems—baggage handling, IT, security, and passenger services—must be integrated and tested in real-world conditions (that is with people and the full end to end process) to ensure seamless functionality.
- **Risk & Contingency Planning**: Identifying potential risks and developing mitigation strategies help prevent disruptions.
- **Relocation & Transition Management**: A structured transition plan balances airline preferences with infrastructure capacity, ensuring minimal operational disruptions during the move.
- **Post-Opening Support & Continuous Improvement**: Ongoing monitoring and issue resolution after launch ensure sustained efficiency and operational excellence.
- **Training & Simulation**: Hands-on training and simulations for airlines, ground handlers, and stakeholders enable smooth transitions from construction to live operations, reducing inefficiencies and enhancing passenger experience.

1 The ORAT methodology has been described in various ways—as Operational Readiness, Activation, and Transition; Operational Readiness and Airport Transfer; or Operational Readiness and Transfer.

Shift to a Proactive Mindset

ORAT is not simply a checklist or a phase that comes after construction; it is a comprehensive approach to establishing a fully operational airport from day one. In the dynamic, high-pressure and interconnected world of aviation, airports are more than just transportation hubs—they are crucial national and global infrastructure. The efficiency and 24/7 operational readiness of an airport are fundamental to its success. However, too many airport projects still focus predominantly on construction, relegating ORAT to an afterthought. The results of this oversight can be disastrous, manifesting in costly delays, system failures and long-term damage to reputation.

A successful ORAT approach is a sequential, phased process that begins during the early design stages and continues through to the airport's operational launch. Rather than waiting until the end of a project, ORAT should be integrated from the very beginning. For instance, the ORAT strategy is ideally developed during the planning and concept design stages of a new terminal, while later phases—such as training and trials—align with the activation stage of a project, after the testing and commissioning. By synchronizing each ORAT phase with design development, every activity occurs at the optimal time, reducing the likelihood of remedial work during testing.

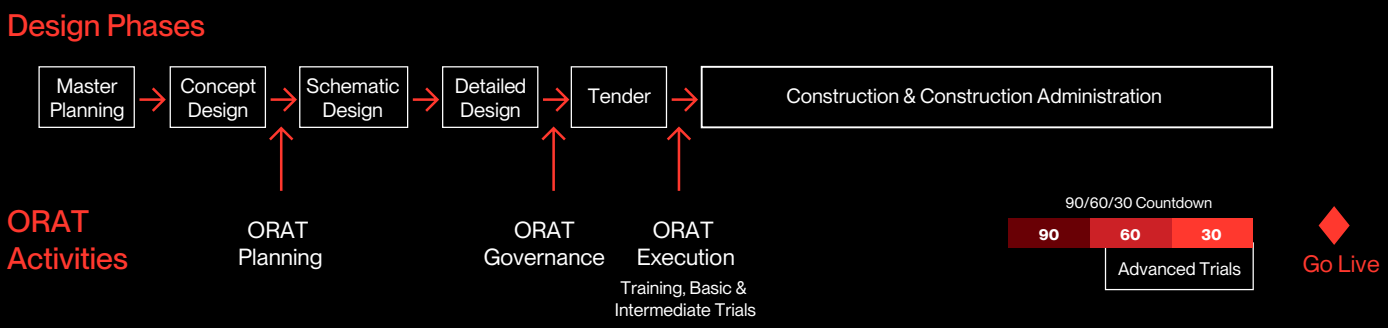


Figure 1 – ORAT Alignment



Why a Sequential Phased Process Works Best

Early Integration – Initiating ORAT in the early planning phases allows for the identification of potential operational issues before they become entrenched, considering and integrating people, processes and technology into the design.

Layered Approach – Overlaying the ORAT process onto the design and development timeline means that each phase of ORAT is addressed in parallel with project milestones, preventing the common mistake of trying to cram all ORAT activities at the end of the project, which can lead to rushed and incomplete testing.

Avoiding Overlap with Commissioning – A frequent error is combining ORAT proving trials with the commissioning phase of construction. Commissioning is typically focused on ensuring that individual systems or pieces of equipment function as designed in isolation. In contrast, ORAT extends beyond this by testing the integrated performance of systems, ensuring that staff are trained and that operational processes are optimized for real-world scenarios. Merging these two phases can produce inaccurate findings and necessitate costly rework.

Cost and Efficiency Benefits – Conducting ORAT activities out of sequence may require adjustments and corrective work based on trial outcomes, which can lead to increased costs and delays. A phased, sequential approach minimizes the risk of resource wastage so that issues are identified and addressed progressively throughout the project lifecycle.

By aligning ORAT with the project's design and development phases, airport operators can achieve a smoother transition to full operational status. This methodical process not only enhances the accuracy of testing but also improves overall operational efficiency and the training of staff, leading to a better experience for both passengers and employees.

Benefits Accrue for Airports and Aviation

The construction of new terminals and runways and the adoption of advanced technologies introduce new operational challenges and thus added complexity for airports to handle. A robust ORAT process helps manage this complexity so that all new infrastructure and systems are integrated effectively into daily operations, minimizing disruptions and supporting the airport's long-term success.

A Shift Toward Holistic Project Management – In the past, airport construction and ORAT were viewed as distinct phases. However, as passenger satisfaction, safety, and operational efficiency have become paramount, ORAT has evolved into an overarching project management framework rather than a standalone phase, increasingly embedded in projects from the early stages of planning. This integrated approach allows for seamless coordination across all project components, mitigating risks and enabling a smooth transition from construction to full-scale operations. As more airports adopt this methodology, consistent improvements in operational performance and passenger experience will reinforce the message that an early start brings success..

Cost Efficiency – While ORAT requires careful planning and resources, it ultimately helps save costs in the long run by preventing costly errors, delays and rework. Testing systems and processes before full-scale operation provides opportunity to identify and resolve issues before they lead to major disruptions, which can be much more expensive to address once the airport is operational. As good experiences accrue based on effective ORAT programs, cost efficiency becomes an understood outcome of the ORAT process, something all airports strive to achieve. ORAT would then become common practice, benefitting the interconnected aviation sector.

Environmental and Safety Considerations – As airports implement sustainable and eco-friendly technologies, ORAT specifies measures to test these innovations so that they work in practice. Whether considering energy-efficient systems or eco-friendly construction materials, ORAT helps validate that all sustainability measures are functional and compliant with environmental standards. Additionally, ORAT's testing phase includes safety drills and contingency planning to make sure that airports meet regulatory requirements for health and safety. Airports everywhere are trying to properly incorporate sustainable technologies, and this objective will continue. Success achieved through one ORAT process can inform another process, helping aviation build knowledge in this area.

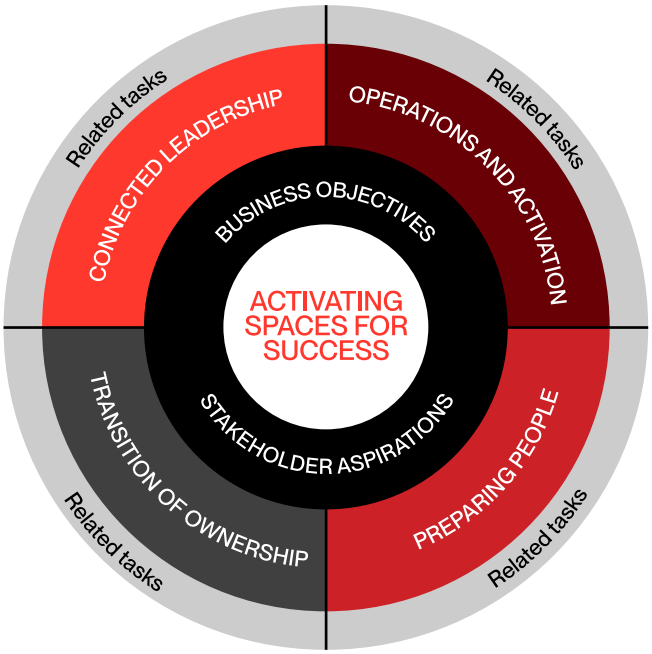


Figure 2 – ORAT Process

Consequences of Not Planning Ahead

Failure to embed ORAT from the outset can lead to a host of issues that are both costly and damaging:

Operational Delays: Without proper integration of ORAT, the delay in opening a facility can be significant. Systems may not be fully tested or integrated, and staff may be inadequately trained. Even a minor delay can result in millions of dollars in lost revenue when considering the daily cost of operations.

Disruption post-opening: Many new airports experience operational chaos post-launch due to incomplete testing and readiness. Frequent system failures, passenger bottlenecks and staffing issues plague operations, leading to passenger dissatisfaction and a tarnished reputation. The reputational fallout from such disruptions can take years to overcome, impacting future traffic and stakeholder confidence.

Reputational Damage: The reputational consequences of a poorly managed airport opening are far-reaching. Even after an airport recovers operationally, the damage to its public image often lingers for years. High-profile examples of delayed or troubled airport openings serve as stark reminders of what can happen when operational readiness is neglected. While these events become cautionary tales, the lasting damage to the airport's credibility can have a long-term impact on future investment and stakeholder trust.

Key steps for a successful execution

Integrated ORAT Strategy

Develop a comprehensive ORAT blueprint that aligns construction, operational teams and technology vendors.

Clearly assign roles and secure regulatory approvals with aviation authorities.

Preparation & Systems Testing

Establish detailed standard operating procedures and implement robust staff training (theoretical and practical).

Conduct integrated systems testing across all critical airport functions and develop contingency plans.

Live Trials & Simulations

Execute full-scale operational trials with real or simulated passengers, including testing check-in, security, baggage handling and boarding.

Run airside and ground handling simulations and perform crisis response exercises to uncover and address deficiencies.

Phased Transition & Ongoing Optimization

Implement a gradual transition from construction to live operations with on-site support and real-time KPI monitoring.

Continue performance reviews and provide post-opening ORAT support to refine processes for operational stability.

Financial Losses: Post-opening disruptions often require expensive retrofits, staff retraining and systems overhaul. These costs are compounded by lost revenue from operational inefficiencies. The financial outlay to recover from an unprepared launch can easily exceed the cost of proactively planning and integrating ORAT.



The Essential Role of ORAT in Future Airport Development

The increasing complexity of airport projects and the growing emphasis on passenger satisfaction make the ORAT process an indispensable part of modern airport management. Beginning with the end in mind through a rigorous ORAT process means that every decision made throughout the lifecycle of the airport project is informed by the need for a smooth transition to live operations—and operational stability.

In today's fast-paced world where travellers have high expectations, ORAT is an essential component of a successful launch. By meticulously planning, testing and preparing all aspects of operations, ORAT enables an efficient and passenger-focused transition from construction to full-scale operations. Without ORAT, the risk of operational failures, delays and reputational damage significantly increases, underscoring its importance for long-term success in modern airport development.

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